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DEPARTMENT OF COMPUTATIONAL SCIENCES
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BACHELOR OF COMPUTER APPLICATIONS (HONOURS) /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS with RESEARCH) - 2024
SEMESTER –V

Sl. No.	Course Code	Course	L	T	P	Evaluation Scheme		Total	Credits	Course Type
						CIA	ESE			
1	BCA50112	Software Engineering	3	1	0	40	60	100	4	Major
2	BCA50113	Design and Analysis of Algorithm	4	0	0	40	60	100	4	Major
3	BCA59114	Design and Analysis of Algorithm Lab	0	0	4	40	60	100	2	Major
4	BCA57115	Full-stack Development -II	2	0	4	40	60	100	4	Major
5	BCA57205	Machine Learning	3	0	2	40	60	100	4	Minor
6	BCA58001	Project-I	0	0	0	40	60	100	2	IAPC
TOTAL			**12**	**1**	**10**	**240**	**360**	**600**	**20**	

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Course Code: BCA50112
Course Name: Software Engineering
Weekly Contact Hours: 3L+1T
Credits: 4
Total Allotted Hours: 45L+15T

Course Objective: The aim of this course is **t**o know and apply basic principles of software engineering to different several applications and to apply the expertise to design, development and testing.

Pre-requisite(s): Knowledge of Software and system development principles.

Course Outcome (COs): After the completion of the course, students would be able to:

CO1: Define the principles of software engineering concept and understand the software development life cycle.
CO2: Explain the significance of different software models and identify their working principles.
CO3: Discuss about knowledge of cost model and its application.
CO4: Compare different testing methodologies and their application scopes.
CO5: Illustrate and criticize the software design strategies and maintenance steps.

Module 1: Introduction to Software Engineering [3H]

Definition of software, types of software, characteristics of software, attributes of good software, define software engineering, software engineering costs, key challenges faced by software engineers, systems engineering concept.

Module 2: Software Development Process Models [10H]

Software process, software process model, the waterfall model, evolutionary development, component-based software engineering (CBSE), process iteration, incremental delivery, spiral development, rapid software development, agile methods, extreme programming, rapid application development (RAD), software prototyping, computer aided software engineering (CASE), overview of CASE approach, classification of CASE tools.

Module 3: Requirement Analysis and Specifications [10H]

Software requirement analysis and specification, system and software requirements, types of software requirements ,functional and non-functional requirements, domain requirements, user requirements, elicitation and analysis of requirements ,overview of techniques, viewpoints interviewing scenarios, use-cases, process modeling with physical and logical DFDs, entity relationship diagram, data dictionary, requirement validation, requirement specification, software requirement specification (SRS), structure and contents SRS format, feasibility.

Module 4: Software Design and Development Tools **[10H]**

Software design concepts, abstraction, architecture, patterns, modularity, cohesion, coupling, information hiding, functional independence, refinement, design of input and control design of user interface design, elements of good design, design issues, features of modern GUI, menus, scroll bars, windows, buttons, icons, panels, error messages etc. coding, programming languages and development tools, selecting languages and tools, good programming practices, coding standards.

Module 5: Software Testing and Project Management **[12H]**

Software testing and quality assurance, verification and validation, different types of testing, design of test cases, quality management activities, product and process, quality standards ISO9000, capability maturity model (CMM),managing software projects, need for the proper management of software, project management activities, project planning, software size estimation and cost estimation, function point analysis ,LOC estimation , COCOMOII model, project scheduling task set for software project, defining a task ,network scheduling, earned value analysis, risk management systems .The RMMM managing people.

Course Outcomes (COs) and Modules Mapping:

Sl. No.	Course outcome (COs)	Mapped Module(s)	Learning levels (as per Bloom's Taxonomy from K1 to K6)
1.	CO1	M1	K1, K2
2.	CO2	M2	K2, K3
3.	CO3	M3	K3, K6
4.	CO4	M4	K3, K4
5.	CO5	M5	K2,K6

Text books:

1. Fundamentals of Software Engineering, Rajib Mall, PHI Learning Pvt. Ltd, 5th Edition, 2018.

Reference books:

1. Software engineering, I. Sommerville, Addison-Wesley Longman,10th Edition,2016.
2. Software engineering: A precise approach, Pankaj Jalote, Wiley-India, 2nd Edition, 2010.
3. Software Engineering: A Practitioner's Approach, Pressman, McGraw Hill Publication, 9th Edition, 2020.

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Course Code: BCA50113
Course Name: Design and Analysis of Algorithm
Weekly Contact Hours: 4L
Credits: 4
Total Allotted Hours: 60L

Course Objective: This course aims to equip students with a comprehensive understanding of algorithm development, implementation and complexity analysis. Topics include stages of algorithm development, time and space complexity analysis, solving recurrences, searching and sorting algorithms, algorithm design techniques graph and tree algorithms, and complexity classes.

Pre-requisite(s): Data-structure and basic programming.

Course Outcomes (COs): After completion of the course students will be able to:

CO1: Employ problem-solving techniques, analyze time and space complexity, asymptotic notations, and solve recurrences using various methods.
CO2: Evaluate and compare linear search, binary search, merge sort, quick sort, and heap sort complexities.
CO3: Apply brute force, implement divide and conquer, illustrate greedy and dynamic programming, implement backtracking techniques.
CO4: Implement graph traversal, analyze shortest path, and evaluate spanning tree algorithms, enhancing skills in solving graph and tree-related problems.
CO5: Acquire insight into computational complexity classes, differentiate P, NP, and NP-complete classes.

Module 1: Algorithm Development & Complexity Analysis **[10H]**

Details of different asymptotic notations for time and space complexity, time-space trade-off, relationships among different asymptotic notations, their mathematical significance. Solving recurrences: substitution method, recurrence tree method and masters' theorem. Finding time complexities by writing programs on a computer.

Module 2: Searching and Sorting Algorithm **[10H]**

Details of Linear Search and Binary search – their comparisons, Interpolation search and its complexity. Merge sort, quick sort and heap sort, their complexity analysis in best case, worst case and average case.

Module 3: Algorithm Design Techniques **[16H]**

Brute force algorithm; divide and conquer strategy - recursive max-min problem, matrix multiplication using divide and conquer and Strassen's algorithm, greedy techniques - fractional knapsack problem, Job sequencing with deadline. Dynamic programming - 0/1 knapsack problem, matrix chain multiplication. Backtracking-N-queens Problem. String matching algorithm – naïve algorithm, KMP algorithm.

Module 4: Graph and Tree Algorithms **[16H]**

Graph traversal algorithms: Depth First Search (DFS) and Breadth First Search (BFS); shortest path algorithms Dijkstra's algorithm, Bellman Ford algorithm, Floyd-Warshall algorithm, disjoined set data structure, spanning tree Prim's and Krushkal algorithm.

Module 5: Complexity Classes **[8H]**

P class, NP class- Boolean satisfiability problem (SAT), Hamiltonian path problem; NP-complete class - Hamiltonian cycle, vertex cover, cook's theorem, basics of approximation algorithms.

Course Outcomes (COs) and Modules Mapping:

Sl. No.	Course Outcome (COs)	Mapped Module(s)	Learning levels (as per Bloom's Taxonomy from K1 to K6)
1.	CO1	M1	K2, K4
2.	CO2	M2	K4, K5
3.	CO3	M3	K2,K3
4.	CO4	M4	K4, K5
5.	CO5	M5	K4,K5

Text books:

1. Introduction to Algorithms, Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2022), MIT Press.

Reference books:

1. Fundamentals of Computer Algorithms, Ellis Horowitz, Sartaj Sahni and S. Rajasekharan, Universities Press, 2nd edition.
2. The Design and Analysis of Computer Algorithms, Aho, A. V., Hopcroft, J. E., & Ullman, J. D. (1974), Addison-Wesley Series, Pearson.

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Course Code: BCA59114
Course Name: Design and Analysis of Algorithm Lab
Weekly Contact Hours: 4P
Credits: 2
Total Allotted Hours: 60P

Course Objective: This course aims to equip students to write programs with different algorithms and implement them to solve real world problems.

Pre-requisite(s): Basic programming knowledge.

Course Outcomes (COs): After completion of the course students will be able to:

CO1: Develop programs to implement recursive functions to solve different problems and analyze the efficiency.
CO2: Develop programs for sorting a given set of elements and analyze its time complexity.
CO3: Solve and analyze the problems using brute force algorithm, greedy strategy, dynamic programming and backtracking strategy.
CO4: Evaluate graph theory algorithms to solve various problems.
CO5: Develop programs on string matching algorithms and analyze them.

Module 1: Recursion **[4H]**

Programs to familiar with recursion-tower of Hanoi, permutations and combinations, recursive maxmin,

Module 2: Searching and Sorting Algorithm **[8H]**

Programs of Linear Search and Binary search – their comparisons, Interpolation search and its complexity. Merge sort, quick sort and heap sort, their complexity analysis in best case, worst case and average case.

Module 3: Algorithm Design Techniques **[24H]**

Writing programs on Brute force algorithm; divide and conquer strategy - matrix multiplication using divide and conquer and Strassen's algorithm, greedy techniques - fractional knapsack problem, Job sequencing with deadline. Dynamic programming - 0/1 knapsack problem, matrix chain multiplication. Backtracking-N-queens Problem.

Module 4: Graph and Tree Algorithms **[20H]**

Programs on Graph traversal algorithms: Depth First Search (DFS) and Breadth First Search (BFS); shortest path algorithms Dijkstra's algorithm, Bellman Ford algorithm, Floyd-Warshall algorithm, spanning tree Prim's and Krushkal algorithm.

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Module 5: String Matching **[4H]**

String matching algorithm – naïve algorithm, KMP algorithm.

Course Outcomes (COs) and Modules Mapping:

Sl. No.	Course Outcome (COs)	Mapped Module(s)	Learning levels (as per Bloom's Taxonomy from K1 to K6)
1.	CO1	M1	K5, K6
2.	CO2	M2	K4, K6
3.	CO3	M3	K4,K5
4.	CO4	M4	K5, K6
5.	CO5	M5	K4, K6

Text books:

1. Introduction to Algorithms, Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2022), MIT Press.

Reference books:

1. Fundamentals of Computer Algorithms, Ellis Horowitz, Sartaj Sahni and S. Rajasekharan, Universities Press, 2nd edition.

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Course Code: BCA57115

Course Name: Full-stack Development-II

Weekly Contact Hours: 2L+4P

Credits: 4

Total Allotted Hours: 30L+ 60P

Course Objective: The aim of this course is to understand the different aspects of AngularJS and ReactJS. Different events, directives and the complete overview of the subject has been covered in this syllabus. The different services, events and deployment will be explored in details.

Pre-requisite(s): Basic knowledge of operating systems and java scripts.

Course Outcome (COs): After the completion of the course, students would be able to:

CO1: Understand and compare the fundamentals of ReactJS and its architecture.
CO2: Discuss and analyze the module, controllers and events.
CO3: Explain and apply react data binding.
CO4: Design and evaluate forms, modules with directives.
CO5: Build and evaluate services on different platforms.

Module 1: React JS and JSX with Component **[4H(L)]**

Introduction to React JS, prerequisites of react JS, why use React JS, software and description, install react JS, creating react app. converting HTML JSX, file and directory structure of react, code snippets, javascript variable in JSX, comments, class component, function component.

Practical: **[8H(P)]**

Installation of react JS, implementation of creating react app. converting HTML JSX, file and directory structure of react, code snippets, javascript variable in JSX, comments, class component, function component.

Module 2: React CSS with Bootstrap package and router concepts **[6H(L)]**

Inline styling, CSS style sheet, CSS module, styled components, adding bootstrap for react, react bootstrap package, react table installation and use. install a react router, add a router in react, concept of props, use of props, passing props to child, create reusable component, props validation, concept of state, state rules, use of state.

Practical: **[12H(P)]**

Implementation of react CSS with Bootstrap and router.

Module 3: User Input Forms with Events and deployment **[6H(L)]**

React arrow function, handling forms, creating event handlers, free react developer tools, deploy react app on server.

Practical: **[12H(P)]**

Implementation of react arrow function, handling forms, creating event handlers, free react developer tools, deploy react app on server.

Module 4: API Request with React and data binding **[6H(L)]**

Fetching data, send data, API create using PHP & MYSQL, introduction to angularJS, MVC architecture, conceptual overview, setting up the environment, first application understanding and attributes, number and string expressions, object binding and expressions, working with arrays forgiving behavior, understanding data binding.

Practical: **[12H(P)]**

Implementation of API and related applications.

Module 5: Working of directives using Controllers with filters and services **[8H(L)]**

Conditional directives, understanding controllers programming controllers & $scope object, adding behavior to a scope object with parameters, nested controllers and scope inheritance, multiple controllers and their scopes, built-in filters, creating custom filter, using simple form, using CSS classes, form events, custom validations, understanding services, developing creating services, using a service, injecting dependencies in a service.

Practical: **[16H(P)]**

Implementation of scope inheritance using controller and use of validation control.

Course Outcomes (COs) and Modules Mapping:

Sl. No.	Course outcome (COs)	Mapped Module(s)	Learning levels (as per Bloom's Taxonomy from K1 to K6)
1.	CO1	M1	K2, K5
2.	CO2	M2	K2, K4
3.	CO3	M3, M4	K2, K3. K5
4.	CO4	M4	K5, K6
5.	CO5	M5	K5, K6

Text books:

1. Learning React: Functional Web Development with React and Redux, Banks, Alex, Porcello, Eve, O'Reilly Media, Inc., 1st Edition, 2017.
2. Complete Book on Angular 4,Nate Murray, Felipe Coury, Ari Lerner and Carlos Taborda , CreateSpace Independent Publis , 1st Edition, 2017.

Reference books:

1. The Complete Book on Angular js, AriLerner, Ng-Book, Lightning Source Inc;1st edition, 2010.

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Course Code: BCA57205
Course Name: Machine Learning
Weekly Contact Hours: 3L+2P
Credits: 4
Total Allotted Hours: 45L + 30P

Course Objective: The main aim of this course is to apply basic principles of AI in solutions that require problem solving, inference, perception, knowledge representation, and learning.

Pre-requisite(s): Knowledge of AI and algorithm.

Course Outcome (COs): After the completion of the course, students would be able to:

CO1: Define Machine Learning and differentiate it with Artificial Intelligence.
CO2: Explain and analyze the significance of using supervised, unsupervised and reinforcement learning.
CO3: Organize and evaluate data preprocessing and training and testing model.
CO4: Relate AI with Machine Learning to solve various business problems.
CO5: Demonstrate, evaluate and develop different Ensemble learning algorithms

Module 1: Introduction to Machine Learning **[8H(L)]**
Understanding machine learning -principle of machine learning, compare machine learning with Artificial Intelligence, defining big data- big data in context with machine learning, leveraging the power of machine learning, descriptive analytics - predictive analytics.

Practical: **[4H(P)]**
Implementation of data visualization and representation using python libraries.

Module 2: Types of Learning and its applications **[12H(L)]**
The roles of statistics and data mining with machine learning, approaches to machine learning, supervised learning, unsupervised learning, reinforcement learning, neural networks. Applying machine learning - understanding machine learning techniques, tying machine learning methods to outcomes, applying machine learning to business needs.

Practical: **[12H(P)]**
Implementation of data preprocessing and machine learning models.

Module 3: Training of Model in machine learning **[10H(L)]**
Looking inside machine learning, the role of algorithms, types of machine learning algorithms, training machine learning systems, data preparation, identify relevant data, the machine learning cycle.

Practical: **[4H(P)]**
Implementation of training algorithms in machine learning model.

Module 4: AI and Machine learning with Business Problems **[10H(L)]**

Understanding how machine learning can help, focus on the business problem, executing a pilot project, determining the best learning model, machine learning skills. Machine learning to provide solutions to business problems, applying machine learning to patient health, proactively responding to IT Issues, protecting against fraud.

Practical: **[6H(P)]**

Implementation of model training and prediction algorithms.

Module 5: Ensemble Learning **[5H(L)]**

Ensemble learning model combination schemes, voting, error-correcting output codes, bagging: random forest trees, boosting: adaboost, stacking.

Practical: **[4H(P)]**

Implementation of ensemble models and evaluations.

Course Outcomes (COs) and Modules Mapping:

Sl. No.	Course outcome (COs)	Mapped Module(s)	Learning levels (as per Bloom's Taxonomy from K1 to K6)
1.	CO1	M1	K1, K4
2.	CO2	M2	K2, K4, K5
3.	CO3	M3	K3, K5
4.	CO4	M4	K2, K6
5.	CO5	M5	K2, K5, K6

Text books:

1. Introduction to Machine Learning Second Edition, The MIT Press Cambridge, Massachusetts, London, England, EthemAlpaydın.

Reference books

1. Hands-On Machine Learning with Scikt-Learn &TensorFlow, O'Reilly Media, AurélienGéron, Inc., 2nd Edition, 2022.
2. Machine Learning Algorithms: Handbook, Clever Fox Publishing, Aman Kharwal ,1st Edition, 2023.

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Course Code: BCA58001
Course Name: Project-I
Weekly Contact Hours: NA
Credits: 2
Total Allotted Hours: NA

Course Objective: The objective of the course is to manage the scope, cost, timing, and quality of the project. Identify project goals, Constraints, deliverables, performance criteria, control needs, and resource requirements in consultation with stakeholders.

Pre-requisite(s): Basic Knowledge of Programming.

Course Outcomes (COs): After the completion of the course, students would be able to:

CO1: Identify & interpret the technical knowledge acquired in the previous semesters for solving real world problems.
CO2: Recap the study of interest and analyze in all aspects.
CO3: Identify the tools and prepare a synopsis for planning of work.